

MAT 120

Beta and Gamma Functions

1. Evaluate in terms of gamma function: (WEEK 2)

$$\begin{aligned} & \text{(i)} \int_0^4 x^{3/2} (4-x)^{5/2} dx, \quad \text{(ii)} \int_0^b y^5 \sqrt{b^2 - y^2} dy, \quad \text{(iii)} \int_0^\infty e^{-t^2} dt, \\ & \text{(iv)} \int_0^\infty x^5 e^{-4x} dx, \quad \text{(v)} \int_0^\infty e^{-y^2} y^5 dy, \quad \text{(vi)} \int_0^\infty e^{-x^2} dx, \quad \text{(vii)} \int_0^\infty x^6 e^{-3x} dx, \\ & \text{(viii)} \int_0^\infty e^{-x^2} x^9 dx, \quad \text{(ix)} \int_0^\infty \sqrt{x} e^{-x^2} dx, \quad \text{(x)} \int_0^1 \frac{x^3}{\sqrt{1-x^3}} dx, \\ & \text{(xi)} \int_0^1 \frac{1}{\sqrt{x \ln(1/x)}} dx, \quad \text{(xii)} \int_0^1 \left(1 - \frac{1}{x}\right)^{1/3} dx. \end{aligned}$$

2. Evaluate in terms of beta function: (WEEK 3)

$$\begin{aligned} & \text{(i)} \int_0^1 \frac{x^2}{\sqrt{1-x}} dx, \quad \text{(ii)} \int_0^1 x^7 (1-x)^3 dx, \quad \text{(iii)} \int_0^1 \frac{1}{\sqrt{1-x^3}} dx, \\ & \text{(iv)} \int_0^1 (1-x)^{1/2} x^3 dx, \quad \text{(v)} \int_0^1 x^{5/2} (1-x)^{3/2} dx, \\ & \text{(vi)} \int_0^a y^7 \sqrt{a^4 - y^4} dy, \quad \text{(vii)} \int_0^4 y^3 \sqrt{64 - y^3} dy, \\ & \text{(viii)} \int_0^1 x^2 (1-x^3)^{3/2} dx, \quad \text{(ix)} \int_0^\infty \frac{1}{1+x^4} dx. \end{aligned}$$

3. Evaluate the following integrals: (WEEK 3)

$$\begin{aligned} & \text{(i)} \int_0^\pi \sin^5 \theta \cos^4 \theta d\theta, \quad \text{(ii)} \int_0^\pi \sin^6 \theta \cos^7 \theta d\theta, \\ & \text{(iii)} \int_0^{\pi/6} \sin^2 6x \cos^4 3x dx, \quad \text{(iv)} \int_0^{\pi/4} \sin^2 4\theta \cos^3 2\theta d\theta, \\ & \text{(v)} \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta, \quad \text{(vi)} \int_0^{\pi/8} \sin^2 8x \cos^4 4x dx. \end{aligned}$$

Formula

1. $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad \text{where } n > 0$
2. $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad \text{where } m > 0, n > 0.$
3. $\int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{2 \Gamma(\frac{p+q+2}{2})}$
4. $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
5. $\Gamma(n) = (n-1)!$
6. $\Gamma(n+1) = n\Gamma(n) = n!$
7. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$
8. $\Gamma(1) = 1$
9. $\Gamma(\frac{p}{2}) = (\frac{p}{2}-1)(\frac{p}{2}-2)(\frac{p}{2}-3)\dots\dots\dots\frac{1}{2} \cdot \Gamma(\frac{1}{2})$
10. $\Gamma(n) = (n-1)(n-2)\dots\dots\dots 3.2.1$
11. $\int_0^{\pi/2} 2 \sin^{2x-1}(t) \cos^{2y-1}(t) dt = \beta(x, y)$

Gamma Beta Distribution Solution

1. (i) $8.4.32 \frac{\Gamma(\frac{5}{2})\Gamma(\frac{7}{2})}{\Gamma(6)}$
 (ii) $\frac{b^7}{2} \frac{\Gamma(3)\Gamma(\frac{3}{2})}{\Gamma(\frac{9}{2})}$
 (iii) $\frac{1}{2} (1/2)$
 (iv) $\frac{1}{4^6} (6)$
 (v) $\frac{1}{2} (3)$
 (vii) $\frac{1}{3^7} (7)$
 (viii) $\frac{1}{2} (5)$
 (ix) $\frac{1}{2} (3/4)$
 (x) $\frac{1}{3} \frac{\Gamma(\frac{4}{3})\Gamma(\frac{1}{3})}{\Gamma(\frac{4}{3}+\frac{1}{3})}$
 (xi) $\sqrt{2} (1/2)$
 (xii) $-\frac{\Gamma(\frac{2}{3})\Gamma(\frac{4}{3})}{\Gamma(2)}$
2. (i) $\beta(3, \frac{1}{2})$
 (ii) $\beta(8, 4)$
 (iii) $\frac{1}{3}\beta(1/3, \frac{1}{2})$
 (iv) $\beta(4, 3/2)$
 (v) $\beta(7/2, 5/2)$
 (vi) $\frac{\alpha^{10}}{4}\beta(2, 3/2)$
 (vii) $4^4 \cdot 8 \cdot \frac{1}{3}\beta(4/3, 3/2)$
 (viii) $\frac{1}{3}\beta(1, 5/2)$
 (ix) $\frac{1}{2} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})}{2\Gamma(1)} = \frac{1}{4} \beta(\frac{1}{4}, \frac{3}{4})$
3. (i) $\frac{\Gamma(3)\Gamma(\frac{5}{2})}{\Gamma(\frac{11}{2})}$ (vi) $\frac{\Gamma(\frac{3}{2})\Gamma(\frac{7}{2})}{2\Gamma(5)}$
 (ii) $\frac{\Gamma(\frac{7}{2})\Gamma(4)}{\Gamma(\frac{15}{2})}$
 (iii) $\frac{4}{3} \frac{\Gamma(\frac{3}{2})\Gamma(\frac{7}{2})}{2\Gamma(5)}$
 (iv) $\frac{\Gamma(\frac{3}{2})\Gamma(3)}{\Gamma(\frac{9}{2})}$
 (v) $\frac{\Gamma(\frac{5}{2})\Gamma(\frac{3}{2})}{2\Gamma(4)}$